PYTHON MENTOR

Automated Code Testing, Defensive Auto-Repair & Interactive Assessment System

Author / Lead: Dhevaesh B N | **GitHub:** https://github.com/Dhevaesh/python-mentor | **Tech:** Pure Python Standard Library + Streamlit

Overview: Python Mentor is an interactive educational assistant and defensive code sandbox designed for students and developers. It stress-tests Python code against boundary conditions, auto-generates defensive input validation guards and try-except blocks, and administers adaptive quizzes based on past errors.



Figure 1: Python Mentor Main Chatbot & Code Workspace Interface

Chatbot Feature	Functionality & Execution
■ **Test Code**	Fuzzes code with edge cases (None, 0, negative values, empty collections) in an isolated 4-second timeout sandbox.
■■ **Fix Code**	Synthesizes input checks (e.g. if n is None or n < 0: return 0) and wraps execution in a defensive try-except block.
■ **Logic Quiz**	Generates interactive multiple-choice questions on loop bounds (< vs <=) and invariants.
■■ **Timed Test**	Assembles timed evaluations with per-question countdown tickers based on past code failures.
■ **Error Notebook**	Durable JSON ledger of runtime exceptions with individual deletion controls and targeted retry prompts.

1. Modular System Architecture (5 Team Members)

The project is structured into 5 independent, cleanly decoupled Python modules (one per teammate) plus a unified web interface:

Member	Module File	Runner (.bat)	Module Responsibility
Member 1	`module_1_test_code.py`	`test_module_1.bat`	**Edge-Case Testing:** Inspects function AST, generates edge inputs, and runs subprocess sandbox.
Member 2	`module_2_fix_code.py`	`test_module_2.bat`	**Defensive Fixer:** Synthesizes boundary guards and wraps function bodies in try-except blocks.
Member 3	`module_3_logic_quiz.py`	`test_module_3.bat`	**Logic Quiz Engine:** Produces dynamic questions evaluating loop boundaries and logic flow.
Member 4	`module_4_timed_test.py`	`test_module_4.bat`	**Timed Assessment:** Builds countdown-driven assessments tracking time per question.
Member 5	`module_5_error_notebook.py`	`test_module_5.bat`	**Error Notebook:** Durable JSON failure ledger with individual item deletion and retry logic.
Lead	`app.py`	`run.bat`	**Platform UI:** Coordinates all modules into an interactive Streamlit web dashboard.

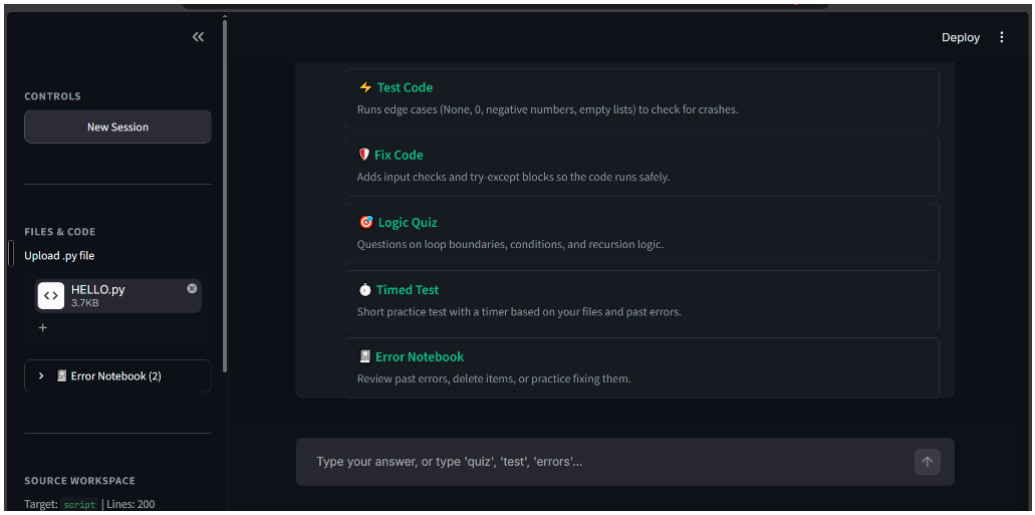


Figure 2: Python Mentor Interactive Chatbot Action Cards

2. Sandbox Execution & Adaptive Logic Quiz

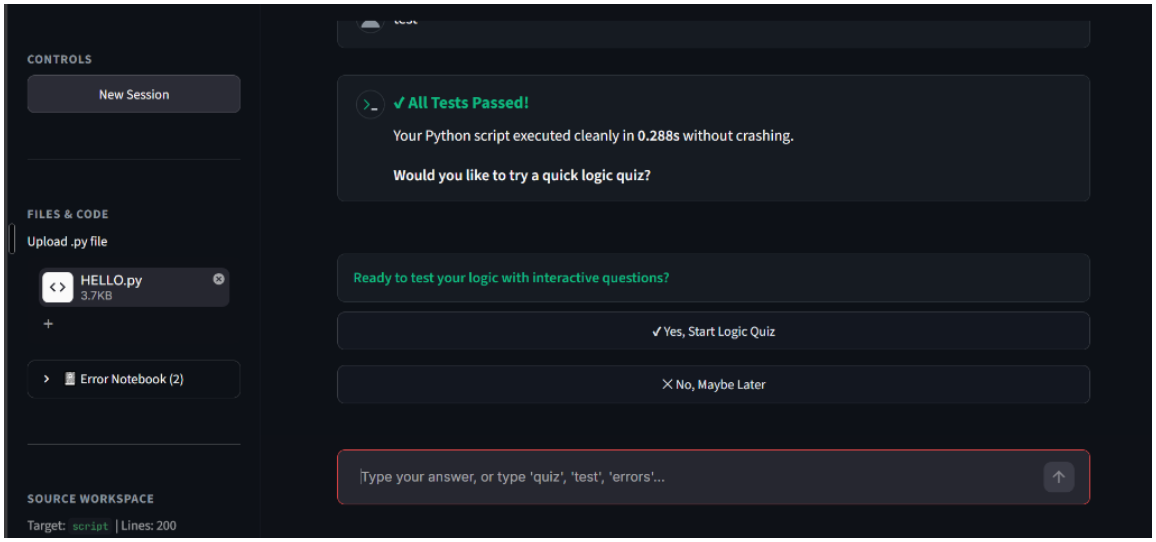


Figure 3: Sandbox Test Execution Result — Code Cleanly Executed in 0.288s with Interactive Prompt

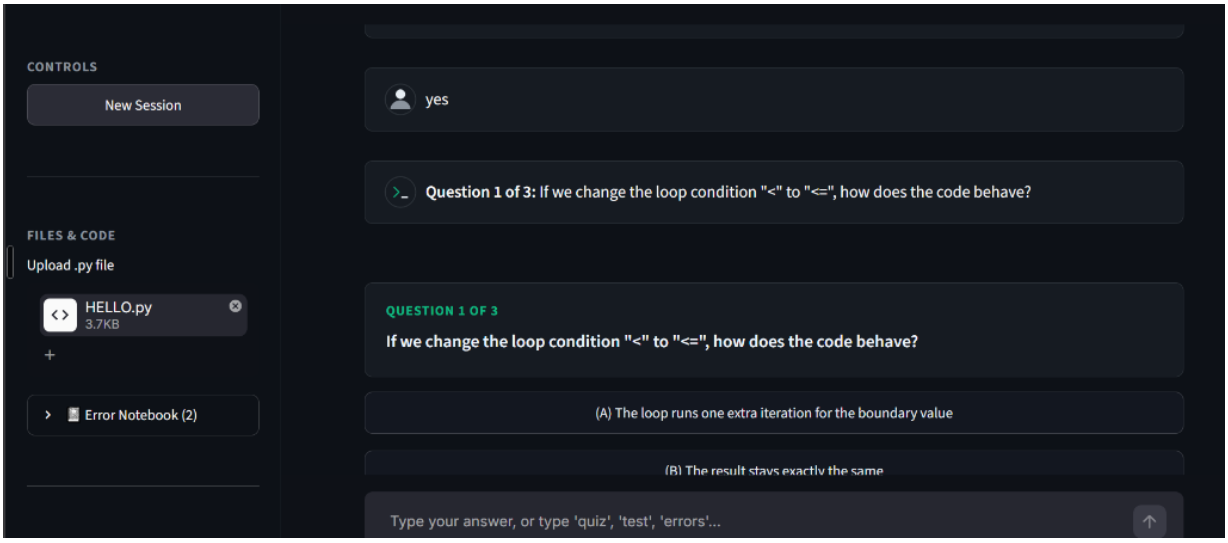


Figure 4: Interactive Logic Quiz — Adaptive Multiple-Choice Assessment with Shuffled Options

3. Defensive Auto-Repair & Error Incident Ledger

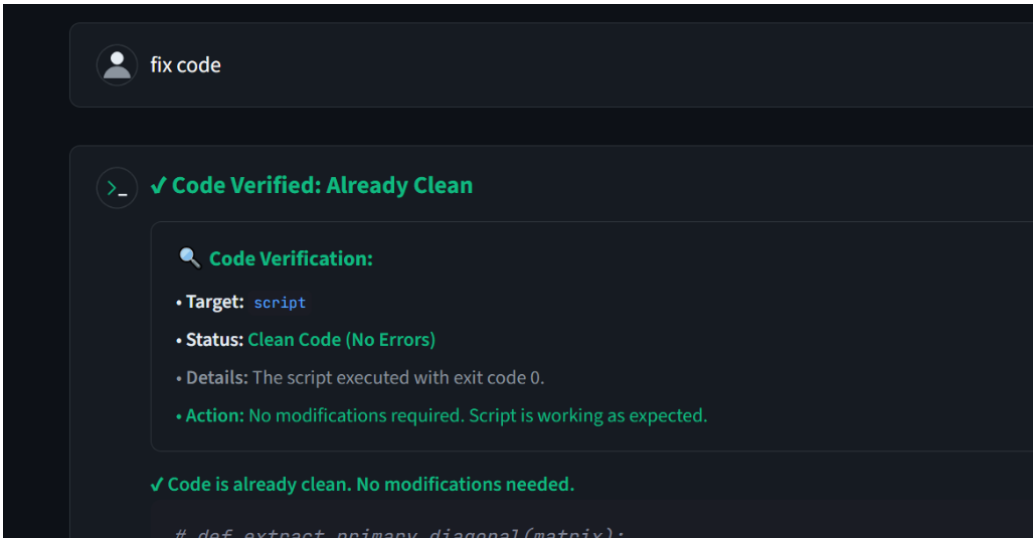


Figure 5: Code Verification & Defensive Patch Analysis — Root Cause Breakdown and Status Card

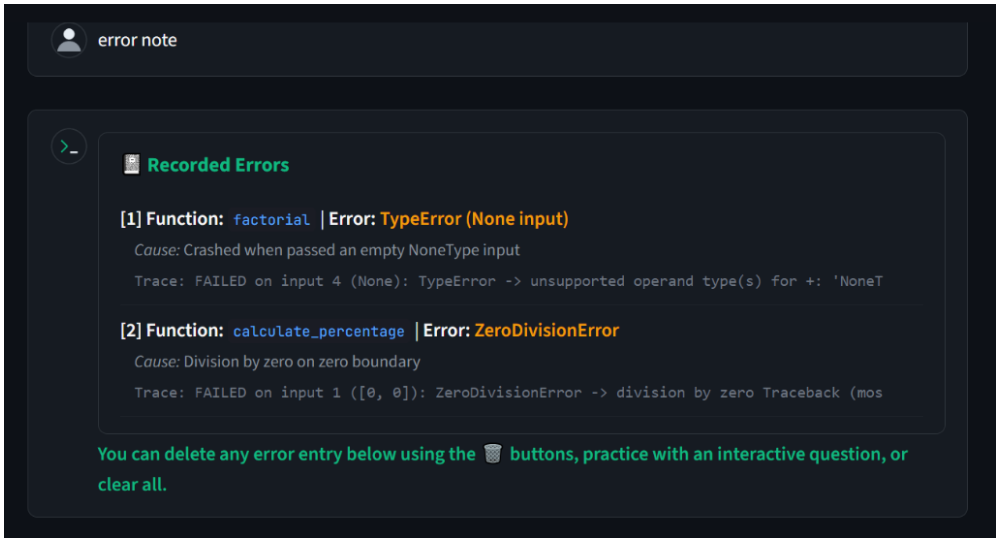


Figure 6: Error Notebook Incident Log — Detailed Incident Classification for TypeError and ZeroDivisionError

4. Granular Error Management & Technical Verification

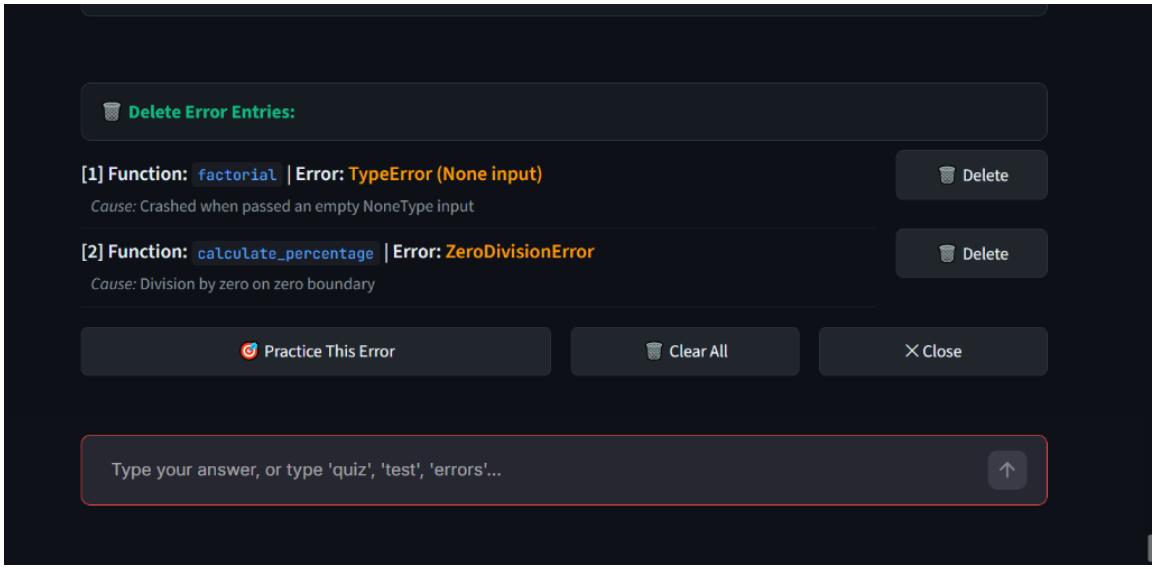


Figure 7: Interactive Error Management Ledger — Individual Deletion Controls, Practice Trigger, and Batch Clear

Key Implementation Highlights

- **100% Pure Python Standard Library:** Zero external AI API calls (no OpenAI, Gemini, or DeepSeek cloud dependencies; zero cost; complete data privacy).
- **Dual Defensive Patching:** Auto-repair adds input parameter boundary checks (if n is None or n < 0: return 0) alongside structured try...except blocks.
- **Subprocess Isolation:** Code execution runs in a sandboxed subprocess with an enforced 4-second timeout to prevent deadlocks and infinite loops.
- **Durable Persistence:** Error history is maintained in user_files/error_notebook.json across restarts and session reloads.

Project Submission & Verification Links

GitHub Repository:	https://github.com/Dhevaesh/python-mentor
Repository Status:	**Public** | Branch: main | All Tests Passing | Zero External Dependencies
Local Project Path:	`C:\Users\dhevb\Desktop\ApexHeal (Launch with run.bat)`
Local Archive:	`C:\Users\dhevb\Desktop\Python_Mentor.zip`